(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3/1, 2 and 3

Ans : (c) Bioremediation uses naturally occurring organisms to break down hazardous substances into less toxic substances. It uses microorganisms to degrade organic contaminations in soil, groundwater, sludge and solids. The microorganisms break down contaminate by using them as an energy source or co-metabolizing them with an energy source.

Not all contaminates are easily treated by bioremediation using microorganisms. For example heavy metals such as cadmium and lead are not readily absorbed or captured by microorganisms.

Genetic engineering has been used to create organisms designed for specific purposes. For e.g. bacterium Deinococcus radiodurans (the most radiation resistant organism known) has been modified to consume highly radioactive nuclear waste.

70. Due to some reasons, if there is a huge fall in the population of species of butterflies, what could be its likely consequence/consequences?

1. Pollination of some plants could be adversely affected.
2. There could be a drastic increase in the fungal infections of some cultivated plants.
3. It could lead to a fall in the population of some species of wasps, spiders and birds.

Select the correct answer using the code given below:

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (c) Statement 1 and 3 are correct. Butterflies are pollinating insects. They help in pollination of many flowering plants. Butterfly also act as a lower member of the food chain. A number of animal, including birds and mice feed on butterfly. (As population of butterfly diminish, so will populations of birds. This is "butterfly effect". Statement 2 is not correct as it does no lead to increase in the fungal infections of some cultivated plants.

71. It is possible to produce algae based biofuels, but what is/are the likely limitation(s) of developing countries in promoting this industry?

1. Production of algae based biofuels is possible in seas only and not on continents.
2. Setting up and engineering the algae based biofuel production requires high level of expertise/technology until the construction is completed.
3. Economically viable production necessitates the setting up of large scale facilities which may raise ecological and social concerns.

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

Ans : (b) Statement 1 is incorrect. Productions of algal biofuel are possible both in seas and on continents. They can grow marginal or non-crop land also on brackish or polluted water. Land based systems are more developed than sea- based systems.

Statement 2 and 3 are correct. Developing and engineering ABB technology requires a high level of expertise until constriction is finished. Innovation for higher productivity also requires the same knowledge and/or experience. All algal based biofuel concepts require significant capital investment. Access to this technology by the poor may be largely difficult. Scale facilities are more economically viable but are also more likely to have higher social and ecological impacts.

72. In the context of mitigating the impending global warming due to anthropogenic emissions of carbon dioxide, which of the following can be the potential sites for carbon sequestration?

1. Abandoned and uneconomic coal seams
2. Depleted oil and gas reservoirs
3. Subterranean deep saline formations

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (d) Carbon sequestration is the process involved in carbon capture and the long-term storage of atmospheric carbon dioxide. Geological sequestration involves the storage of CO_2 underground in depleted oil and gas reservoirs, saline formations or deep, unmixable coal beds.

73. The term 'Domestic Content Requirement' is sometimes seen in the news with reference to

(a) Developing solar power production in our country
(b) Granting licenses to foreign T.V. channels in our country
(c) Exporting our food products to other countries
(d) Permitting foreign educational institutions to set up their campuses in our country

Ans : (a) The term 'Domestic Content Requirement' is sometimes seen in the news with reference to developing solar power production in our country.

- It was instituted in the Jawaharalal Nehru National solar mission

National Solar Mission-

- The National Solar Mission is also known as the Jawharlal Nehru National Solar Mission.
- It was adopted by India in 2010.
- It aims to target the generation of 100,000 MW of grid connected Solar power capacity by 2022.
- A mandatory domestic content requirement was imposed on solar power developers participating in phase I and phases II.

74. According to the Wildlife (Protection) Act, 1972, which of the following animals cannot be hunted by any person except under some provisions provided by law?

1. Gharial
2. Indian wild ass
3. Wild buffalo

Select the correct answer using the code given below:

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (d) Gharial (Gavialis Gangeticus), Indian wild ass (Equus hemionus khur) and wild Buffalo (Bubalus bubalis) are all maintained under schedule 1 for the wildlife (Protection) Act, 1972.

75. Consider the following statements:

1. Climate and Clean Air Coalition (CCAC) to Reduce Short Lived Climate Pollutants is a unique initiative of G20 group of countries.
2. The CCAC focuses on methane, black carbon and hydro fluorocarbons.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans : (b) The Climate and Clean Air Coalition to reduce short lived climate pollutants (CCAC) was launched in 2012. The government of Bangladesh, Canada, Ghana Mexico, Sweden and the United states, along with the United Nation & Environment Programme (UNEP), came together to initiate the first effort to treat these pollutants as a collective challenge. Statement 1 is incorrect.

The coalitions initial focus is on methane, black carbon and HFCs. Statement 2 is correct.

CURRENT AFFAIRS

76. What is the purpose of 'Vidyanjali Yojana'?

1. To enable the famous foreign educational institutions to open their campuses in India.
2. To increase the quality of education provided in government schools by taking help from the private sector and the community.
3. To encourage voluntary monetary contributions from private individuals and organizations so as to improve the infrastructure facilities for primary and secondary schools.

Select the correct answer using the code given below:

(a) 2 only (b) 3 only
(c) 1 and 2 only (d) 2 and 3 only

Ans : (a) The ministry of HRD recently launched the 'Vidyanjali Scheme aimed at boosting the education system by delivering volunteer teachers to government schools. It will not replace the regular and professionally qualified teachers in the government schools. The volunteer's responsibility is towards the overall development of the child, not academics. The volunteer service will be used in developing skills like public speaking, creative writing counselling, music and dance.

77. What is the aim of the programme 'Unnat Bharat Abhiyan'?

(a) Achieving 100% literacy by promoting collaboration between voluntary organizations and government's education system and local communities.
(b) Connecting institutions of higher education with local communities to address development challenges through appropriate technologies.
(c) Strengthening India's scientific research institutions in order to make India a scientific and technological power.
(d) Developing human capital by allocating special funds for health care and education of rural and urban poor, and organizing skill development programmes and vocational training for them.

Ans : (b) With a view to uplift rural India, the Unnat Bharat Abhiyan programme is launched in collaboration with the Indian Institutes of Technology (IIT) and the National Institutes of Technology (NITs) across the country.

The programme aims to connect institutions of higher education, including Indian Institutes of Technology (IITs), National Institute of Technology (NITs), and Indian Institutes of Science Education & Research (IISERs) etc, with local communities to address the development challenges through appropriate technologies.

78. The term 'Digital Single Market Strategy' seen in the news refers to

(a) ASEAN (b) BRICS
(c) EU (d) G 20

Ans : (c) The Digital Single Market Strategy aims to open up digital opportunities for people and businesses and enhance Europe's position as a world leader in the digital economy. The Digital Single Market is part of the Digital Agenda for Europe 2020 program of the EU an initiative of Europe 2020 proposed strategy. It is defined by a Digital Single Market Strategy for Europe by European Commission.

79. In India, it is legally mandatory for which of the following to report on cyber security incidents?

1. Service providers
2. Data centres
3. Body corporate

Select the correct answer using the code given below:

(a) 1 only (b) 1 and 2 only
(c) 3 only (d) 1, 2 and 3